

Joint Exclusivity

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Abstract

We introduce *joint exclusivity* (JE), a flexible extension of mutual exclusivity for non-negative random vectors that prohibits only the simultaneous positivity of all n components. Unlike mutual exclusivity, JE admits a broad class of marginals while retaining a sharp existence criterion. We prove that a JE random vector with prescribed marginals $F_1, \dots, F_n$ exists if and only if

$$\sum_{i=1}^n \bar{F}_i(0) \leq n - 1.$$

We also provide a canonical face-based construction with freely specified copulas within each face, and show that the trivariate case can elegantly be reduced to a one-parameter family. Finally, we extend the framework through Wang-type distortions and an m -exclusivity hierarchy interpolating between mutual exclusivity and JE.

Key words and phrases: counter-monotonicity; joint exclusivity; mutual exclusivity; m -exclusivity; generalised joint exclusivity; joint mixability; distortion functions.

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1 Introduction

Dependence modelling plays a central role in quantitative risk analysis, where the choice of joint distribution profoundly shapes aggregate loss behaviour and the values of associated risk measures (Cheung et al., 2014; Deelstra et al., 2011; Dhaene et al., 2002a; Lauzier et al., 2023). Among all dependence structures, those of an extremal nature are of special importance: they yield sharp bounds on risk measures and serve as natural benchmarks for sensitivity analysis.

In the bivariate setting, the positive extreme is *co-monotonicity* (Dhaene et al., 2002b; Nelsen, 2010), which captures the notion that both components always move in the same direction. Formally, $X_{12} = (X_1, X_2)$ is co-monotonic if

$$X_{12} \stackrel{d}{=} (h_1(Z), h_2(Z)),$$

for some random variable Z and functions h_1, h_2 that are both non-decreasing (or both non-increasing). This structure is characterised by the Fréchet upper bound

$$F_{12}(x_1, x_2) = \min \{F_1(x_1), F_2(x_2)\},$$

where F_{12} denotes the joint cumulative distribution function (CDF) of X_{12} and F_1, F_2 are the marginal distributions. Co-monotonicity represents the strongest form of positive dependence: an increase in one component almost surely does not decrease the other.

At the opposite extreme lies *counter-monotonicity* (Dhaene and Denuit, 1999; Nelsen, 2010), which captures perfect negative association. A bivariate vector X_{12} is counter-monotonic if

$$X_{12} \stackrel{d}{=} (h_1(Z), h_2(Z)),$$

where h_1 is non-decreasing and h_2 is non-increasing (or vice versa), for some random variable

Z . Its joint distribution is given by the Fréchet lower bound

$$F_{12}(x_1, x_2) = \max \{F_1(x_1) + F_2(x_2) - 1, 0\}.$$

In the bivariate case, both Fréchet bounds define valid joint CDFs for arbitrary marginals F_1 and F_2 , so the two extremal dependence structures are uniquely and unambiguously defined.

These two notions, however, behave very differently in higher dimensions. Let $N = \{1, \dots, n\}$ denote the index set. Co-monotonicity extends naturally to the n -variate setting: for $X_N = (X_1, \dots, X_n)$, the representation

$$X_N \stackrel{d}{=} (h_1(Z), \dots, h_n(Z)),$$

where all h_i are non-decreasing (or all non-increasing), remains well-defined for any $n \geq 2$. Moreover, the Fréchet upper bound

$$F_N(x_1, \dots, x_n) = \min_{i \in N} F_i(x_i)$$

continues to define a valid joint distribution for arbitrary marginals and any $n \geq 2$.

Counter-monotonicity, by contrast, does not admit a canonical multivariate extension. For $n \geq 3$, if two pairs of components are each driven by opposing monotone functions of a common variable, the induced relationship between the remaining pair is necessarily co-monotone, yielding intricacies that are handled non-uniquely. Correspondingly, the multivariate Fréchet lower bound

$$F_N(x_1, \dots, x_n) = \max \left\{ \sum_{i \in N} F_i(x_i) - (n - 1), 0 \right\}$$

fails, in general, to define a valid joint distribution when $n \geq 3$. As a result, several extensions of counter-monotonicity have been proposed in the literature, each tailored to specific modelling

contexts.

Among the strongest of these is *mutual exclusivity* (ME) (Cheung and Lo, 2014; Dhaene and Denuit, 1999). Letting $\bar{F}$ denote the decumulative distribution function (DDF), a non-negative random vector X_N is ME if

$$\bar{F}_{ij}(0, 0) = 0, \quad \text{for all } i < j, \quad i, j \in N, \quad (1.1)$$

that is, no two components are simultaneously positive. An ME random vector with prescribed marginals $F_1, \dots, F_n$ exists if and only if

$$\sum_{i \in N} \bar{F}_i(0) \leq 1. \quad (1.2)$$

Whenever (1.2) holds, the Fréchet lower bound is a valid joint distribution and coincides with that of the ME random vector. This tractability is one of the principal appeals of ME: existence is fully determined by the single inequality (1.2), and the joint distribution is directly given by the Fréchet lower bound. However, condition (1.2) is rather restrictive—it requires that the aggregate probability of any component being strictly positive does not exceed unity—which considerably limits the practical applicability of ME.

Alternative notions of counter-monotonicity have been introduced to overcome this limitation, including joint mixability (Wang and Wang, 2016), d -counter-monotonicity (Lee and Ahn, 2014), and Σ -counter-monotonicity; see Puccetti and Wang (2015) for a unified treatment. These approaches typically impose that a prescribed function of the components is almost surely constant, thereby permitting a broader class of marginals. However, they are often difficult to construct explicitly or to characterise completely for a given set of marginals. This stands in sharp contrast to ME, whose existence is fully determined by condition (1.2) and whose joint distribution is directly given by the Fréchet lower bound. This observation motivates the follow-

ing question: *does there exist a notion of counter-monotonicity that is more flexible than ME, yet remains amenable to explicit construction and admits a complete, tractable characterisation analogous to (1.2)?*

A natural answer emerges by relaxing the pairwise requirement in (1.1). Rather than prohibiting any pair of components from being simultaneously positive, one may instead require only that all n components are never simultaneously positive. This leads to the following definition.

Definition 1.1. *A non-negative random vector X_N is said to be jointly exclusive (JE) if*

$$\bar{F}_N(0, \dots, 0) = 0, \text{ or equivalently, } \prod_{i \in N} X_i = 0 \text{ almost surely,} \quad (1.3)$$

that is, its support is contained in $[0, \infty)^n \setminus (0, \infty)^n$.

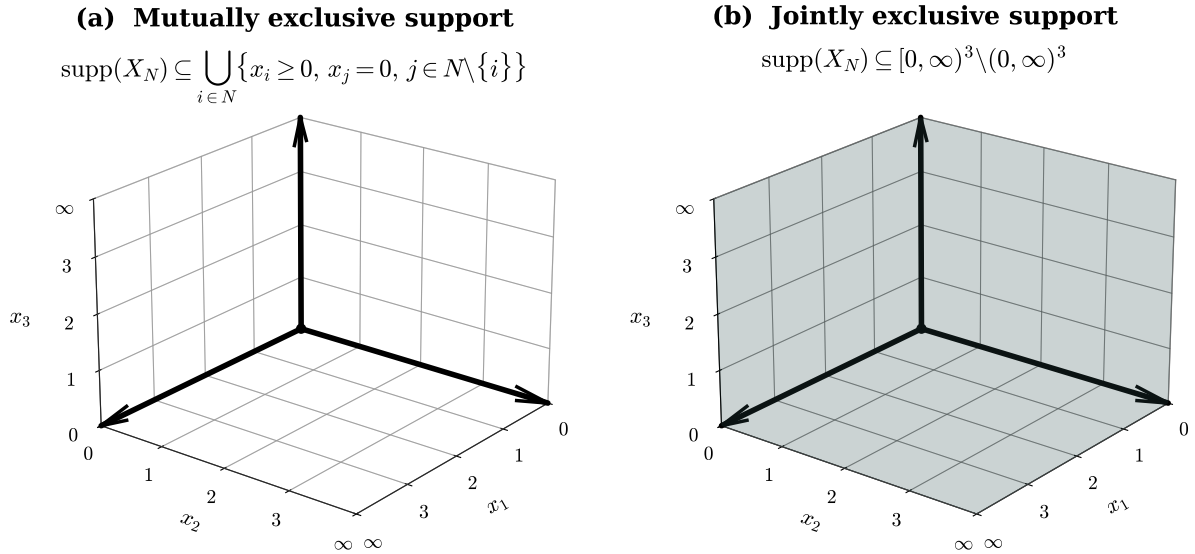


Figure 1: Comparison of the supports of ME and JE random vectors for $n = 3$.

When $n = 2$, conditions (1.1) and (1.3) coincide, so ME and JE are equivalent. For $n \geq 3$, ME implies JE but not conversely, making JE a strictly more flexible notion; see Figure 1 for

an illustration of the distinction between the two supports when $n = 3$. As we demonstrate in this paper, JE additionally admits a canonical construction and a complete characterisation of the admissible marginals, thereby providing an affirmative answer to the question above.

The remainder of the paper is organised as follows. Section 2 introduces the notation and presents basic properties of JE. Section 3 develops the main construction methods and characterisation results. Further remarks on the reflected and translated variants of JE as well as its connection to joint mixability are discussed in Section 4. Section 5 concludes.

2 Preliminaries

Throughout the paper we work with the index set $N = \{1, \dots, n\}$. Define $M = \{I \subset N : 2 \leq |I| \leq n - 1\}$ as the collection of all proper subsets of N of intermediate cardinality, and for each $i \in N$ let $M_i = \{I \in M : i \in I\}$ denote those containing i ; the latter family arises naturally in the constructions of Section 3. We primarily consider non-negative random vectors $X_N = (X_1, \dots, X_n)$, with sub-vectors $X_I = (X_i, i \in I)$ for $I \subseteq N$, and write $x_N = (x_1, \dots, x_n) \in \mathbb{R}^n$ and $x_I = (x_i, i \in I)$ for their deterministic counterparts. We use F , $\bar{F}$, f , and φ to denote the CDF, DDF, density function, and characteristic function, respectively, with subscripts identifying the corresponding vector or sub-vector; for example, F_N and $\bar{F}_N$ denote the CDF and DDF of X_N , and analogously for F_I and $\bar{F}_I$.

By Definition 1.1 and as illustrated in Figure 1, the support of a JE random vector is contained in $\bigcup_{i \in N} \{x \in [0, \infty)^n : x_i = 0\}$, a union of n coordinate hyperplanes of dimension $n - 1$ within $[0, \infty)^n$. The joint distribution therefore assigns no n -dimensional Lebesgue mass and is entirely characterised by its lower-dimensional marginals, as formalised below.

Proposition 2.1. *A non-negative random vector X_N is JE if and only if its CDF satisfies*

$$F_N(x_N) = \begin{cases} 1 - \sum_{i \in N} \bar{F}_i(x_i) + \sum_{I \in M} (-1)^{|I|} \bar{F}_I(x_I) \\ \quad = (-1)^{n-1} \left(1 - \sum_{i \in N} F_i(x_i) + \sum_{I \in M} (-1)^{|I|} F_I(x_I) \right), & x_N \in [0, \infty)^n, \\ 0, & \text{otherwise,} \end{cases} \quad (2.1)$$

and if and only if its characteristic function satisfies

$$\varphi_N(t_N) = (-1)^{n-1} \left(1 - \sum_{i \in N} \varphi_i(t_i) + \sum_{I \in M} (-1)^{|I|} \varphi_I(t_I) \right), \quad t_N \in \mathbb{R}^n. \quad (2.2)$$

Proof. CDF representation. For $x_N \in [0, \infty)^n$, the inclusion–exclusion principle yields

$$F_N(x_N) = \sum_{J \subseteq N} (-1)^{|J|} \bar{F}_J(x_J), \quad \bar{F}_\emptyset \equiv 1. \quad (2.3)$$

Under JE, non-negativity gives $\bar{F}_N(x_N) = \mathbb{P}(X_i > x_i, \forall i \in N) \leq \mathbb{P}(\prod_{i \in N} X_i > 0) = 0$ for all $x_N \in [0, \infty)^n$. Dropping the $J = N$ term from (2.3) yields the first expression in (2.1). Conversely, if F_N satisfies this expression, note that it equals $\sum_{J \subseteq N} (-1)^{|J|} \bar{F}_J(x_J)$; substituting into the right-hand side of (2.3) then gives $(-1)^n \bar{F}_N(x_N) = 0$, confirming JE. For $x_N \notin [0, \infty)^n$, $F_N(x_N) = 0$ follows immediately from non-negativity.

For the second expression, the dual inclusion–exclusion identity $\bar{F}_N(x_N) = \sum_{J \subseteq N} (-1)^{|J|} F_J(x_J)$, combined with $\bar{F}_N(x_N) = 0$, gives $(-1)^n F_N(x_N) = -\sum_{J \subseteq N} (-1)^{|J|} F_J(x_J)$, from which $F_N(x_N) = (-1)^{n-1} \sum_{J \subseteq N} (-1)^{|J|} F_J(x_J)$, yielding the second expression.

Characteristic function representation. Under JE, $\prod_{i \in N} X_i = 0$ a.s., so for any $t_N \in \mathbb{R}^n$ at least one factor satisfies $e^{it_i \cdot 0} - 1 = 0$, giving $\prod_{i \in N} (e^{it_i X_i} - 1) = 0$ a.s. Expanding via the identity $\prod_{i \in N} (c_i - 1) = \sum_{J \subseteq N} (-1)^{n-|J|} \prod_{j \in J} c_j$ and taking expectations yields $\sum_{J \subseteq N} (-1)^{n-|J|} \varphi_J(t_J) = 0$. Isolating the $J = N$ term and rearranging gives (2.2). The converse follows from the

uniqueness theorem for characteristic functions together with the CDF equivalence established above. $\square$

Remark 2.2. *The representation (2.1) illuminates the structural distinction between JE and ME. Under ME, no two components are simultaneously positive, so $\bar{F}_I(x_I) = 0$ for every $I \in M$ and $x_I \in [0, \infty)^{|I|}$. The interaction terms*

$$\sum_{I \in M} (-1)^{|I|} \bar{F}_I(x_I) \tag{2.4}$$

therefore vanish identically, and the CDF reduces to

$$F_N(x_N) = \begin{cases} 1 - \sum_{i \in N} \bar{F}_i(x_i) = \sum_{i \in N} F_i(x_i) - (n - 1), & x_N \in [0, \infty)^n, \\ 0, & \text{otherwise,} \end{cases}$$

which coincides with the Fréchet lower bound and defines a valid CDF under condition (1.2). Under JE, by contrast, the interaction terms (2.4) need not vanish, reflecting the richer dependence structure that permits components to be simultaneously positive in pairs or smaller sub-groups while still satisfying the global JE constraint.

3 Main Results

We begin by establishing the fundamental characterisation of JE in terms of the admissible marginal distributions.

Theorem 3.1. *Let $F_1, \dots, F_n$ be marginal distributions on $[0, \infty)$. A JE random vector X_N with marginals $F_1, \dots, F_n$ exists if and only if*

$$\sum_{i \in N} \bar{F}_i(0) \leq n - 1. \tag{3.1}$$

Proof. Necessity. Suppose that X_N is JE with marginals $F_1, \dots, F_n$. By Definition 1.1, $\bar{F}_N(0, \dots, 0) = 0$. The Fréchet lower bound for the joint DDF gives

$$0 \leq \max \left\{ \sum_{i \in N} \bar{F}_i(0) - (n - 1), 0 \right\} \leq \bar{F}_N(0, \dots, 0) = 0,$$

from which $\sum_{i \in N} \bar{F}_i(0) \leq n - 1$ follows immediately.

Sufficiency. Assume that (3.1) holds. We construct a valid JE distribution with marginals $F_1, \dots, F_n$ by assigning probability mass to each piece of the partition

$$[0, \infty)^n \setminus (0, \infty)^n = \{0_N\} \cup \bigcup_{i \in N} A_i \cup \bigcup_{I \in M} E_I,$$

where 0_N denotes the origin, $A_i = \{x_N : x_i > 0, x_j = 0 \ \forall j \neq i\}$ is the i -th open semi-axis, and

$$E_I = \{x_N : x_i > 0 \ \forall i \in I, x_j = 0 \ \forall j \in N \setminus I\}$$

is the open face indexed by $I \in M$. These sets are mutually disjoint.

Step 1: Intermediate faces. For each $I \in M$ and $x_I \in [0, \infty)^{|I|}$, set

$$\mathbb{P}(X_i > x_i, i \in I; X_j = 0, j \in N \setminus I) = p_I C_I \left(\frac{\bar{F}_i(x_i)}{\bar{F}_i(0)}, i \in I \right), \quad (3.2)$$

where $p_I \geq 0$ is the total probability mass assigned to face E_I and C_I is an arbitrary copula on $[0, 1]^{|I|}$ (Nelsen, 2010). The argument $\frac{\bar{F}_i(x_i)}{\bar{F}_i(0)} \in [0, 1]$ is the survival function of X_i conditional on $X_i > 0$; when $\bar{F}_i(0) = 0$ for some $i \in I$, the constraint (3.4) forces $p_I = 0$, so the formula need only be evaluated when all denominators are strictly positive. Since the faces $\{E_I : I \in M\}$ are disjoint, the copulas $\{C_I : I \in M\}$ may be chosen independently; in particular, C_{I_1} need not be a marginal of C_{I_2} even when $I_1 \subset I_2$.

Step 2: Coordinate axes. The marginal constraint $\mathbb{P}(X_i > x_i) = \bar{F}_i(x_i)$ must hold for each $i \in N$. The contribution of face E_I (for $I \ni i$) to $\mathbb{P}(X_i > x_i)$ is obtained by setting $x_k = 0$ (and hence $\frac{\bar{F}_k(x_k)}{\bar{F}_k(0)} = 1$) for all $k \in I \setminus \{i\}$ in (3.2), which by the copula marginalisation property $C(u, 1, \dots, 1) = u$ yields $p_I \cdot \frac{\bar{F}_i(x_i)}{\bar{F}_i(0)}$. Summing these face contributions over all $I \in M_i$ and requiring equality with $\bar{F}_i(x_i)$ uniquely determines the axis mass:

$$\mathbb{P}(X_i > x_i; X_j = 0, j \in N \setminus \{i\}) = \bar{F}_i(x_i) \left(1 - \frac{\sum_{I \in M_i} p_I}{\bar{F}_i(0)} \right), \quad x_i \in [0, \infty). \quad (3.3)$$

This expression defines a valid (non-negative, non-increasing) survival function for the i -th axis component if and only if

$$\sum_{I \in M_i} p_I \leq \bar{F}_i(0), \quad i \in N. \quad (3.4)$$

Step 3: Origin. The total probability mass on all axes equals $\sum_{i \in N} [\bar{F}_i(0) - \sum_{I \in M_i} p_I] = \sum_{i \in N} \bar{F}_i(0) - \sum_{I \in M} |I| p_I$, and the total mass on all faces equals $\sum_{I \in M} p_I$. Subtracting from unity, the origin receives mass

$$\mathbb{P}(X_i = 0, i \in N) = 1 - \sum_{i \in N} \bar{F}_i(0) + \sum_{I \in M} (|I| - 1) p_I, \quad (3.5)$$

which is non-negative if and only if

$$\sum_{I \in M} (|I| - 1) p_I \geq \sum_{i \in N} \bar{F}_i(0) - 1. \quad (3.6)$$

One can verify that the total probability sums to unity under this assignment. The construction therefore yields a valid JE distribution if and only if there exist coefficients $p_I \geq 0$ satisfying both (3.4) and (3.6).

Step 4: Feasibility via linear programming. The question reduces to whether the maximum of

$\sum_{I \in M} (|I| - 1)p_I$ over the feasible set defined by (3.4) is at least $\sum_{i \in N} \bar{F}_i(0) - 1$. Consider the primal linear program

$$\begin{aligned} & \text{maximise} && \sum_{I \in M} (|I| - 1)p_I, \\ & \text{subject to} && p_I \geq 0, \quad \sum_{I \in M_i} p_I \leq \bar{F}_i(0), \quad i \in N, \end{aligned} \tag{3.7}$$

with dual problem

$$\begin{aligned} & \text{minimise} && \sum_{i \in N} \bar{F}_i(0) r_i, \\ & \text{subject to} && r_i \geq 0, \quad \sum_{i \in I} r_i \geq |I| - 1, \quad I \in M. \end{aligned} \tag{3.8}$$

Both programs are feasible: $p_I = 0$ is primal feasible, and setting $r_i = 1$ for all i is dual feasible, since $|I| \geq |I| - 1$ for every $I \in M$. By strong duality, the optimal values of (3.7) and (3.8) coincide, so the sufficiency condition becomes

$$\min_{r \text{ dual-feasible}} \sum_{i \in N} \bar{F}_i(0) r_i \geq \sum_{i \in N} \bar{F}_i(0) - 1.$$

The dual minimum is attained at a vertex of the feasible polytope. An analysis of the extreme points of the dual polytope reveals that the minimum of the dual objective is achieved either at the symmetric point $r^* = (\frac{n-2}{n-1}, \dots, \frac{n-2}{n-1})$ or at one of the n permutations of $(1, \dots, 1, 0)$, with the zero in position j . Feasibility of both candidates is immediate: at r^* , $\sum_{i \in I} r_i^* = |I| \cdot \frac{n-2}{n-1} \geq |I| - 1$ holds for all $I \in M$ because $|I| \leq n - 1$; at $(1, \dots, 1, 0_j, 1, \dots, 1)$, the dual constraint evaluates to $|I|$ or $|I| - 1$ according to whether $j \notin I$ or $j \in I$, both of which satisfy $\geq |I| - 1$. The corresponding objective values are $\frac{n-2}{n-1} \sum_{i \in N} \bar{F}_i(0)$ and $\sum_{i \in N} \bar{F}_i(0) - \bar{F}_j(0)$;

minimising the latter over j gives $\sum_{i \in N} \bar{F}_i(0) - \max_{i \in N} \bar{F}_i(0)$. Hence

$$\min_{r \text{ dual-feasible}} \sum_{i \in N} \bar{F}_i(0) r_i = \min \left\{ \frac{n-2}{n-1} \sum_{i \in N} \bar{F}_i(0), \sum_{i \in N} \bar{F}_i(0) - \max_{i \in N} \bar{F}_i(0) \right\}. \quad (3.9)$$

It remains to verify that this minimum is at least $\sum_{i \in N} \bar{F}_i(0) - 1$ as (3.6) dictates. If the minimum in (3.9) is attained by the second term, the requirement reduces to $\max_{i \in N} \bar{F}_i(0) \leq 1$, which holds trivially since each $\bar{F}_i(0)$ is a probability. If it is attained by the first term, the requirement becomes

$$\frac{n-2}{n-1} \sum_{i \in N} \bar{F}_i(0) \geq \sum_{i \in N} \bar{F}_i(0) - 1,$$

which simplifies to $\sum_{i \in N} \bar{F}_i(0) \leq n-1$, precisely condition (3.1). In both cases, (3.1) is sufficient to guarantee the existence of feasible coefficients p_I , and the construction in Steps 1–3 defines a valid JE random vector with the prescribed marginals $F_1, \dots, F_n$. $\square$

The necessary and sufficient condition (3.1) for JE is significantly less restrictive than its ME counterpart (1.2), and the gap widens with increasing dimension n . Equivalently in terms of the marginal CDFs, the JE condition reads

$$\sum_{i \in N} F_i(0) \geq 1,$$

while ME requires the substantially stronger

$$\sum_{i \in N} F_i(0) \geq n-1.$$

Under ME, the constraint $\sum_{i \in N} \bar{F}_i(0) \leq 1$ precludes any marginal from being free of an atom at zero unless all other components are almost surely zero: if $\bar{F}_{i_0}(0) = 1$ for some $i_0 \in N$, then $\sum_{j \neq i_0} \bar{F}_j(0) \leq 0$, forcing $X_j = 0$ almost surely for every $j \neq i_0$. Hence, for non-degenerate

marginals, ME requires $F_i(0) > 0$ for every $i \in N$. Under JE, by contrast, few marginals need carry atoms at zero. If $n - 2$ marginals are fully supported on $(0, \infty)$, each contributes $\bar{F}_i(0) = 1$, so condition (3.1) reduces to

$$\bar{F}_{i_1}(0) + \bar{F}_{i_2}(0) \leq 1, \quad \text{equivalently} \quad F_{i_1}(0) + F_{i_2}(0) \geq 1,$$

for the two remaining marginals. Hence, as few as two marginals need carry atoms at zero, subject to the mild joint condition that their point masses at zero sum to at least one, while the remaining $n - 2$ marginals may be fully supported on $(0, \infty)$.

To further investigate the structural properties of JE, we adopt the construction introduced in the proof of Theorem 3.1 as a canonical model.

Definition 3.2. *A random vector X_N is called a canonical JE random vector if its distribution is specified by (3.2), (3.3), and (3.5) for some coefficients $p_I \in [0, 1]$, $I \in M$, satisfying (3.4) and (3.6).*

For fixed marginals $F_1, \dots, F_n$, if condition (3.1) holds, Theorem 3.1 guarantees the existence of a canonical JE random vector. Its CDF admits the following explicit form, as a direct consequence of Proposition 2.1.

Corollary 3.3. *A random vector X_N is canonical JE if and only if its joint CDF is given by*

$$F_N(x_N) = \begin{cases} 1 - \sum_{i \in N} \bar{F}_i(x_i) + \sum_{I \in M} (-1)^{|I|} \sum_{\substack{J \in M \\ I \subseteq J}} p_J C_J \left(\frac{\bar{F}_i(x_i)}{\bar{F}_i(0)}, i \in I; 1, j \in J \setminus I \right), & x_N \in [0, \infty)^n, \\ 0, & \text{otherwise.} \end{cases} \quad (3.10)$$

Proof. For any $I \in M$ and $x_I \in [0, \infty)^{|I|}$, the canonical construction concentrates mass for the

event $\{X_i > x_i, i \in I\}$ exclusively on faces G_J with $J \supseteq I$, so that

$$\bar{F}_I(x_I) = \sum_{\substack{J \in M \\ J \supseteq I}} p_J C_J \left(\frac{\bar{F}_i(x_i)}{\bar{F}_i(0)}, i \in I; 1, j \in J \setminus I \right),$$

where the arguments equal one for $j \in J \setminus I$ because $x_j = 0$ implies $\bar{F}_j(x_j)/\bar{F}_j(0) = 1$, by the copula marginalisation property $C(u, 1, \dots, 1) = u$. Substituting this expression into the first line of (2.1) yields (3.10). Conversely, given a CDF of the form (3.10), the face probabilities $\{p_J\}$ and copulas $\{C_J\}$ are recoverable from the joint DDFs $\bar{F}_I$, confirming that the random vector is canonical JE. $\square$

For canonical JE random vectors, the coefficients p_I determine how probability mass is allocated across the faces of the support, and thereby indirectly along the axes and at the origin. The admissible values of $\{p_I\}_{I \in M}$ are governed entirely by the marginal quantities $\bar{F}_1(0), \dots, \bar{F}_n(0)$, as encoded in conditions (3.4) and (3.6). In contrast, the copulas C_I control the internal dependence structure within each face E_I ; they may be chosen independently across different $I \in M$ and are not constrained by the marginals.

The following remark illustrates several instructive choices of the parameters p_I and their implications for the resulting dependence structure.

Remark 3.4.

- (1) If $p_I = 0$ for some $I \in M$, then face E_I is excluded from the support. Moreover, the Fréchet lower bound applied to (3.2) at $x_I = 0$ gives

$$\max \left\{ \sum_{i \in I} \bar{F}_i(0) + \sum_{j \in N \setminus I} (1 - \bar{F}_j(0)) - (n - 1), 0 \right\} \leq p_I = 0,$$

which yields the necessary condition

$$\sum_{i \in I} \bar{F}_i(0) - \sum_{j \in N \setminus I} \bar{F}_j(0) \leq |I| - 1.$$

At the extreme, if $p_I = 0$ for all $I \in M$, all probability mass is concentrated on the axes and the origin, and no two components are simultaneously positive. This recovers the ME structure, and condition (3.6) reduces to

$$\sum_{i \in N} \bar{F}_i(0) \leq 1,$$

which is precisely the ME condition (1.2).

- (2) If $p_I = 1$ for some $I \in M$, all probability mass is concentrated on face E_I , and necessarily $p_J = 0$ for all $J \in M \setminus \{I\}$. The Fréchet upper bound applied to (3.2) at $x_I = 0$ requires

$$1 = p_I \leq \min \left\{ \min_{i \in I} \bar{F}_i(0), \min_{j \in N \setminus I} (1 - \bar{F}_j(0)) \right\},$$

which implies $\bar{F}_i(0) = 1$ (i.e. $\mathbb{P}(X_i > 0) = 1$) for all $i \in I$, and $\bar{F}_j(0) = 0$ (i.e. $X_j = 0$ almost surely) for all $j \in N \setminus I$.

- (3) The mass on the i -th axis is given by (3.3). Zero axis mass for component i requires equality in (3.4): $\sum_{I \in M_i} p_I = \bar{F}_i(0)$. Imposing this simultaneously for all $i \in N$ yields a system of n linear equations in $|M| = 2^n - n - 2$ unknowns. For $n \geq 4$, the system is under-determined and generally admits infinitely many solutions.

For $n = 3$, the system is square ($|M| = 3$ equations in 3 unknowns) and has the unique solution

$$p_I = \frac{\sum_{i \in I} \bar{F}_i(0) - \bar{F}_j(0)}{2}, \quad j \in N \setminus I, \quad I \in M.$$

This solution satisfies $p_I \geq 0$ for all $I \in M$ if and only if $\max_{k \in N} \bar{F}_k(0) \leq \sum_{j \neq k} \bar{F}_j(0)$ for every $k \in N$, that is, the three quantities $\bar{F}_1(0), \bar{F}_2(0), \bar{F}_3(0)$ satisfy the triangle inequality. When this condition fails, simultaneous zero axis mass across all three axes is infeasible.

- (4) The mass at the origin is governed by (3.5). Provided that (3.4) holds, assigning zero mass to the origin requires equality in (3.6):

$$\sum_{I \in M} (|I| - 1) p_I = \sum_{i \in N} \bar{F}_i(0) - 1.$$

- (5) Suppose $\sum_{i \in N} \bar{F}_i(0) = n - 1$, so that equality is achieved in (3.1). Let $c_0 = \mathbb{P}(X_i = 0, i \in N)$ denote the origin mass and $c_i = \mathbb{P}(X_i > 0; X_j = 0, j \in N \setminus \{i\})$ the mass on the i -th axis. Summing (3.3) over all $i \in N$ and evaluating at $x_i = 0$ gives

$$\sum_{i \in N} c_i + \sum_{I \in M} |I| p_I = n - 1. \quad (3.11)$$

Since $|I| \leq n - 1$ for every $I \in M$, and the total face mass satisfies $\sum_{I \in M} p_I = 1 - c_0 - \sum_{i \in N} c_i$ by normalization, we have

$$\sum_{I \in M} |I| p_I \leq (n - 1) \sum_{I \in M} p_I = (n - 1) \left(1 - c_0 - \sum_{i \in N} c_i \right).$$

Substituting into (3.11) and rearranging yields

$$(n - 1) c_0 + (n - 2) \sum_{i \in N} c_i \leq 0.$$

Since $n \geq 2$ implies $n - 1 \geq 1$ and $n - 2 \geq 0$, and all terms are non-negative, it follows that $c_0 = 0$ and $c_i = 0$ for every $i \in N$. Consequently, when equality holds in (3.1), the JE random vector concentrates its mass entirely on the faces, with $\sum_{I \in M} p_I = 1$.

Once the marginals are fixed, the admissible coefficients $\{p_I\}_{I \in M}$ are precisely those satisfying (3.4) and (3.6). In general, these constraints define a polytope in $\mathbb{R}^{|M|}$, leaving substantial freedom in the choice of dependence structure. For $n = 3$, however, this polytope has a particularly tractable structure. As noted in Remark 3.4-(3), the sum in (3.6) equals one half of the aggregate of the three sums in (3.4), rendering one constraint redundant. The system then reduces to three independent linear constraints in three unknowns, and the feasible set collapses from a three-dimensional region to a one-dimensional affine family. In particular, all three coefficients p_I can be jointly parametrised by a single scalar $\lambda \in [0, 1]$, interpolating linearly between their extremal values implied by (3.4) and (3.6). The following proposition makes this structure explicit.

Proposition 3.5. *Let $n = 3$ and suppose that condition (3.1) holds. For each $I \in M$, define*

$$p_I(\lambda) = \lambda \min_{i \in I} \bar{F}_i(0) + (1 - \lambda) \max \left\{ \sum_{i \in I} \bar{F}_i(0) - 1, 0 \right\}, \quad \lambda \in [0, 1]. \quad (3.12)$$

Then there exists $\lambda^ \in [0, 1]$ such that the coefficients $\{p_I(\lambda^*)\}_{I \in M}$ satisfy (3.4) and (3.6).*

Proof. Assume (3.1) and define, for each $I \in M$,

$$U_I = \min_{i \in I} \bar{F}_i(0), \quad L_I = \max \left\{ \sum_{i \in I} \bar{F}_i(0) - 1, 0 \right\}.$$

These are the Fréchet upper and lower bounds for p_I : U_I is the maximum value consistent with (3.4) individually, and L_I is the minimum imposed by the bivariate Fréchet lower bound applied to (3.2). The inequality $U_I \geq L_I$ holds because $\min_{i \in I} \bar{F}_i(0) \geq \sum_{i \in I} \bar{F}_i(0) - 1$ reduces to $\max_{i \in I} \bar{F}_i(0) \leq 1$, which is immediate.

Substituting the linear interpolation $p_I(\lambda) = \lambda U_I + (1 - \lambda) L_I$ into (3.4) and (3.6) and solving for λ yields the requirement

$$\ell \leq \lambda \leq u_i \quad \text{for all } i \in N, \quad (3.13)$$

where

$$\ell = \max \left\{ \frac{\sum_{i \in N} \bar{F}_i(0) - 1 - \sum_{I \in M} L_I}{\sum_{I \in M} (U_I - L_I)}, 0 \right\}, \quad u_i = \frac{\bar{F}_i(0) - \sum_{I \in M_i} L_I}{\sum_{I \in M_i} (U_I - L_I)}.$$

We show that $\ell \leq u_i \leq 1$ for every $i \in N$, so that a valid λ^* exists.

Non-negativity of u_i . For $n = 3$ and each $i \in N$, the two sets $I \in M_i$ each have $|I| = 2$, so $L_I = \max\{\bar{F}_i(0) + \bar{F}_k(0) - 1, 0\}$ for the other element $k \in I \setminus \{i\}$. The numerator $\bar{F}_i(0) - \sum_{I \in M_i} L_I$ takes one of the following values according to how many L_I are positive: $\bar{F}_i(0)$, $1 - \bar{F}_k(0)$ for $k \in N \setminus \{i\}$, or $2 - \sum_{j \in N} \bar{F}_j(0)$. Each of these is non-negative: the first two trivially, and the last by condition (3.1). Hence $u_i \geq 0$.

Upper bound $u_i \leq 1$. The bound $u_i \leq 1$ is equivalent to $\bar{F}_i(0) \leq \sum_{I \in M_i} U_I$. For $n = 3$ and fixed i , $\sum_{I \in M_i} U_I = \min(\bar{F}_i(0), \bar{F}_j(0)) + \min(\bar{F}_i(0), \bar{F}_k(0))$ for $\{j, k\} = N \setminus \{i\}$. This sum falls below $\bar{F}_i(0)$ only when $\bar{F}_i(0) > \bar{F}_j(0) + \bar{F}_k(0)$, i.e. when $\bar{F}_i(0)$ strictly dominates both others. Since $\bar{F}_j(0) > \bar{F}_i(0)$ and $\bar{F}_i(0) > \bar{F}_k(0) + \bar{F}_j(0)$ cannot hold simultaneously, the condition $\sum_{I \in M_i} U_I < \bar{F}_i(0)$ fails for at most one index i . The minimum $\min_{i \in N} u_i$ is therefore governed by the remaining indices, for which $u_i \leq 1$, so $\min_{i \in N} u_i \leq 1$.

Interval non-emptiness: $\ell \leq \min_{i \in N} u_i$. Since $\sum_{I \in M_i} (U_I - L_I) \leq \sum_{I \in M} (U_I - L_I)$ and the numerator of u_i is non-negative,

$$u_i \geq \frac{\bar{F}_i(0) - \sum_{I \in M_i} L_I}{\sum_{I \in M} (U_I - L_I)}.$$

Using $\sum_{j \in N \setminus \{i\}} \bar{F}_j(0) - 1 - L_{N \setminus \{i\}} \leq 0$ (a consequence of $L_{N \setminus \{i\}} \geq \sum_{j \neq i} \bar{F}_j(0) - 1$) and $\sum_{I \in M} L_I = \sum_{I \in M_i} L_I + L_{N \setminus \{i\}}$, we obtain

$$u_i \geq \frac{\bar{F}_i(0) - \sum_{I \in M_i} L_I + \sum_{j \in N \setminus \{i\}} \bar{F}_j(0) - 1 - L_{N \setminus \{i\}}}{\sum_{I \in M} (U_I - L_I)} = \frac{\sum_{j \in N} \bar{F}_j(0) - 1 - \sum_{I \in M} L_I}{\sum_{I \in M} (U_I - L_I)}.$$

Since this lower bound is precisely $\max\{\cdot, 0\}$ -truncated to give ℓ , we have $u_i \geq \ell$ for every

$i \in N$. Therefore, λ^* exists in $[\ell, \min_{i \in N} u_i] \subseteq [0, 1]$. $\square$

Example 3.6. Let $n = 3$, so that $M = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$. Suppose $\bar{F}_i(x_i) = \frac{1}{2}(1 - x_i)$ for $x_i \in [0, 1]$ and $i \in N$, corresponding to $X_i \sim \text{Uniform}[0, 1]$ marginals. Then $\sum_{i \in N} \bar{F}_i(0) = 3/2$, so condition (3.1) is satisfied. Applying Proposition 3.5 with $U_I = \min_{i \in I} \bar{F}_i(0) = 1/2$ and $L_I = \max\{\sum_{i \in I} \bar{F}_i(0) - 1, 0\} = 0$ for every $I \in M$ gives

$$p_I(\lambda) = \frac{\lambda}{2}, \quad \lambda \in \left[\frac{1}{3}, \frac{1}{2}\right].$$

At $\lambda = 1/3$, equality holds in (3.6), so the mass at the origin vanishes (see Remark 3.4-(4)). At $\lambda = 1/2$, equality holds in (3.4) for every i , so all axes carry zero mass (see Remark 3.4-(3)). Intermediate values of λ distribute mass across both the faces and the axes.

Now choose the copulas

$$C_{12}(u_1, u_2) = u_1 u_2, \quad C_{13}(u_1, u_3) = \min\{u_1, u_3\}, \quad C_{23}(u_2, u_3) = \max\{u_2 + u_3 - 1, 0\},$$

so that the three faces exhibit, respectively, independence, co-monotonicity, and counter-monotonicity.

By Corollary 3.3 and the identity $\bar{F}_i(x_i)/\bar{F}_i(0) = 1 - x_i$ for $x_i \in [0, 1]$, the joint CDF is

$$F_N(x_N) = \begin{cases} 1 - \frac{1}{2} \sum_{i \in N} (1 - x_i) + \frac{\lambda}{2} (1 - x_1)(1 - x_2) \\ \quad + \frac{\lambda}{2} \min\{1 - x_1, 1 - x_3\} + \frac{\lambda}{2} \max\{1 - x_2 - x_3, 0\}, & x_N \in [0, 1]^3, \\ 0, & \text{otherwise.} \end{cases}$$

The support of X_N is illustrated in Figure 2.

The Pearson correlations are computed by observing that, on face E_I , $X_i|_{\text{face}} \sim \text{Uniform}[0, 1]$, and contributions from all faces not containing both components vanish. This gives $\mathbb{E}[X_i] = 1/4$, $\text{Var}(X_i) = 5/48$, and cross-moments $\mathbb{E}[X_1 X_2] = \lambda/8$ (independence), $\mathbb{E}[X_1 X_3] = \lambda/6$ (co-

Support of canonical JE X_N

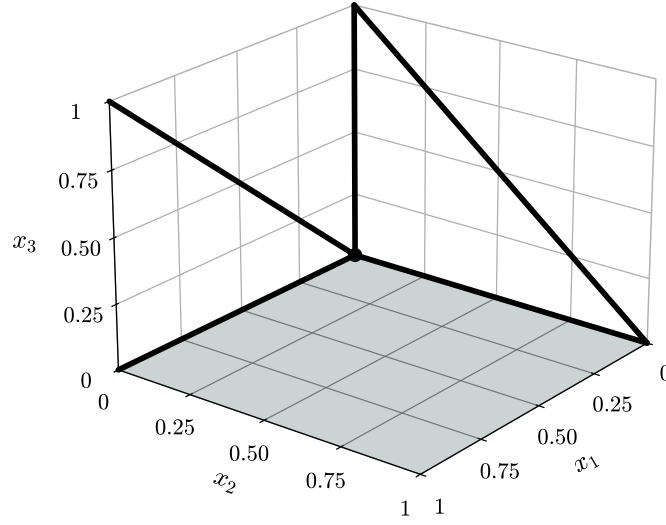


Figure 2: Support of the canonical JE random vector X_N with copulas C_{12} , C_{13} , and C_{23} .

monotone, $X_1 = X_3$ a.s. on face $E_{\{1,3\}}$, $\mathbb{E}[X_2 X_3] = \lambda/12$ (counter-monotone, $X_2 + X_3 = 1$ a.s. on face $E_{\{2,3\}}$), yielding

$$\rho_{12} = \frac{6\lambda - 3}{5}, \quad \rho_{13} = \frac{8\lambda - 3}{5}, \quad \rho_{23} = \frac{4\lambda - 3}{5}.$$

For example,

$$\begin{aligned} \lambda = \frac{1}{3} : \quad & \rho_{12} = -\frac{1}{5}, \quad \rho_{13} = -\frac{1}{15}, \quad \rho_{23} = -\frac{1}{3}, \\ \lambda = \frac{5}{12} : \quad & \rho_{12} = -\frac{1}{10}, \quad \rho_{13} = \frac{1}{15}, \quad \rho_{23} = -\frac{4}{15}, \\ \lambda = \frac{1}{2} : \quad & \rho_{12} = 0, \quad \rho_{13} = \frac{1}{5}, \quad \rho_{23} = -\frac{1}{5}. \end{aligned}$$

All three correlations increase linearly in λ , reflecting the diminishing influence of axis mass (which induces strong negative dependence through the mutual exclusion effect) as face mass grows. At $\lambda = 1/2$, all mass is on the faces and the correlations are determined entirely by the

copulas C_I .

This example highlights the flexibility of the canonical JE framework. Although the global structure enforces a form of counter-monotonicity through the JE constraint, the model accommodates a wide range of local dependence structures through the joint specification of the copulas C_I and coefficients p_I .

Although the canonical JE framework covers a wide range of modelling choices, the dependence structure within each face E_I is constrained by the marginals through the ratios $\bar{F}_i(x_i)/\bar{F}_i(0)$. The copulas C_I may be chosen freely, but the distributional mass on each face is spread over the range of the original marginals. To afford greater control over the effective range of the face distribution, we introduce a generalization that replaces the ratio $\bar{F}_i(x_i)/\bar{F}_i(0)$ by a prescribed transformation $G_i(\bar{F}_i(x_i))$. We call the resulting class the *generalised JE* or G-JE.

Definition 3.7. *For each $i \in N$, let $G_i : [0, 1] \rightarrow [0, 1]$ be a non-decreasing, left-continuous function satisfying $G_i(u) = 0$ for $u \in [0, a_i]$ and $G_i(u) = 1$ for $u \in [b_i, 1]$, where $0 \leq a_i < b_i \leq \bar{F}_i(0)$. The boundary condition $b_i \leq \bar{F}_i(0)$ ensures $G_i(\bar{F}_i(0)) = 1$, which guarantees that the face probability at $x_I = 0$ equals p_I and that the origin mass formula is unchanged. The composition $G_i \circ \bar{F}_i$ is right-continuous in x_i since $\bar{F}_i$ is right-continuous and non-increasing, and left-continuity of G_i ensures $G_i(\bar{F}_i(x_n)) \rightarrow G_i(\bar{F}_i(x))$ as $x_n \downarrow x$.*

A random vector X_N is said to be G-JE if, for each $I \in M$,

$$\mathbb{P}(X_i > x_i, i \in I; X_j = 0, j \in N \setminus I) = p_I C_I(G_i(\bar{F}_i(x_i)), i \in I), \quad x_i \in [0, \infty)^{|I|}; \quad (3.14)$$

for each $i \in N$,

$$\mathbb{P}(X_i > x_i; X_j = 0, j \in N \setminus \{i\}) = \bar{F}_i(x_i) - \left(\sum_{I \in M_i} p_I \right) G_i(\bar{F}_i(x_i)), \quad x_i \in [0, \infty); \quad (3.15)$$

and the origin probability is as in (3.5). The coefficients $p_I \geq 0$ satisfy condition (3.6) together with

$$\sum_{I \in M_i} p_I \leq G_i^*, \quad i \in N, \quad (3.16)$$

where

$$G_i^* = \inf_{\substack{0 \leq x_i < y_i \\ G_i(\bar{F}_i(x_i)) > G_i(\bar{F}_i(y_i))}} \frac{\bar{F}_i(x_i) - \bar{F}_i(y_i)}{G_i(\bar{F}_i(x_i)) - G_i(\bar{F}_i(y_i))}. \quad (3.17)$$

The quantity G_i^* is the smallest local ratio of the decrease in $\bar{F}_i$ to the corresponding decrease in $G_i \circ \bar{F}_i$; it controls how much face mass the i -th axis can support while remaining a valid non-negative, non-increasing function of x_i . The canonical JE is recovered as the special case $G_i(u) = u/\bar{F}_i(0)$, for which $G_i^* = \bar{F}_i(0)$ and condition (3.16) reduces to (3.4).

Theorem 3.8. *Let $F_1, \dots, F_n$ be marginal distributions on $[0, \infty)$ and let $G_1, \dots, G_n$ be functions as in Definition 3.7. A G -JE random vector X_N with marginals $F_1, \dots, F_n$ exists if and only if*

$$\sum_{i \in N} \bar{F}_i(0) - 1 \leq \min \left\{ \frac{n-2}{n-1} \sum_{i \in N} G_i^*, \sum_{i \in N} G_i^* - \max_{i \in N} G_i^* \right\}. \quad (3.18)$$

Proof. Validity of the construction. Since each G_i is left-continuous with $G_i(u) = 0$ for $u \leq a_i$ and $G_i(\bar{F}_i(0)) = 1$, the face probabilities in (3.14) define valid DDFs on E_I : at $x_I = 0$ they give $p_I C_I(1, \dots, 1) = p_I$, and they are non-increasing in each x_i and right-continuous.

Axis probabilities. Define $h_i(u) = u - s_i G_i(u)$ for $u \in [0, \bar{F}_i(0)]$, where $s_i = \sum_{I \in M_i} p_I$. The axis probability for component i equals $h_i(\bar{F}_i(x_i))$. For this to define a valid (non-negative, non-increasing in x_i) survival function on the i -th axis, h_i must be non-decreasing as a function of u on $[0, \bar{F}_i(0)]$. Since $G_i(0) = 0$, we have $h_i(0) = 0$; non-decreasingness of h_i then ensures non-negativity. For any $u > v$ with $G_i(u) > G_i(v)$, non-decreasingness requires $u - v \geq s_i(G_i(u) - G_i(v))$, i.e. $s_i \leq (u - v)/(G_i(u) - G_i(v))$. Taking the infimum over all such pairs yields condition (3.16), with G_i^* as defined in (3.17). Moreover, since $G_i(\bar{F}_i(0)) = 1$, taking $y_i \rightarrow \infty$ in (3.17) shows $G_i^* \leq \bar{F}_i(0)$, ensuring non-negativity also at $x_i = 0$.

Origin probability. Since $G_i(\bar{F}_i(0)) = 1$ for each i , the total mass on the axes equals $\sum_{i \in N} [\bar{F}_i(0) - s_i]$ and the total face mass is $\sum_{I \in M} p_I$. Subtracting from unity gives the origin mass $1 - \sum_{i \in N} \bar{F}_i(0) + \sum_{I \in M} (|I| - 1)p_I$, which coincides with (3.5) and is non-negative if and only if (3.6) holds.

Necessity and sufficiency of (3.18). The construction yields a valid G-JE distribution if and only if there exist $p_I \geq 0$ simultaneously satisfying (3.16) and (3.6). Applying the same linear programming argument as in (3.7)–(3.8), with G_i^* in place of $\bar{F}_i(0)$, the primal maximum of $\sum_{I \in M} (|I| - 1)p_I$ subject to $p_I \geq 0$ and (3.16) equals, by strong duality,

$$\min \left\{ \frac{n-2}{n-1} \sum_{i \in N} G_i^*, \sum_{i \in N} G_i^* - \max_{i \in N} G_i^* \right\}.$$

Feasible p_I satisfying (3.6) exist if and only if this maximum is at least $\sum_{i \in N} \bar{F}_i(0) - 1$, which is precisely condition (3.18). $\square$

Remark 3.9.

(1) *The G-JE is strictly more flexible than canonical JE: the parameters a_i and b_i restrict the effective support on each face E_I to a sub-range of the marginal support, while the functions G_i introduce distributional distortions within that range, all without disrupting the global JE constraint. The cost is a more stringent existence condition (3.18), owing to the inequality $G_i^* \leq \bar{F}_i(0)$ established in the proof of Theorem 3.8. When $\bar{F}_i$ attains both boundary values a_i and b_i , this bound sharpens further: if there exist $x_i^{(1)} < x_i^{(2)}$ such that $\bar{F}_i(x_i^{(1)}) = b_i$ and $\bar{F}_i(x_i^{(2)}) = a_i$, then*

$$G_i^* \leq \frac{b_i - a_i}{G_i(b_i) - G_i(a_i)} = b_i - a_i \leq \bar{F}_i(0),$$

so the admissible range for $\{p_I\}_{I \in M}$ is correspondingly smaller than in the canonical case.

(2) Suppose $\bar{F}_i$ is continuous at some $x_i^* \in (0, \infty)$ while $G_i \circ \bar{F}_i$ has a left-jump there, i.e. $c := \lim_{x \nearrow x_i^*} G_i(\bar{F}_i(x)) - G_i(\bar{F}_i(x_i^*)) > 0$. Since G_i is left-continuous and $\bar{F}_i$ is right-continuous, such a left-jump in $G_i \circ \bar{F}_i$ arises precisely when G_i has a right-discontinuity at $\bar{F}_i(x_i^*)$: as $x_i^{(k)} \nearrow x_i^*$, one has $\bar{F}_i(x_i^{(k)}) \searrow \bar{F}_i(x_i^*)$ from above, so $G_i(\bar{F}_i(x_i^{(k)}))$ converges to the right-limit $G_i(\bar{F}_i(x_i^*)^+) = G_i(\bar{F}_i(x_i^*)) + c$. Taking $y_i = x_i^*$ and $x_i = x_i^{(k)} \nearrow x_i^*$ in (3.17): continuity of $\bar{F}_i$ gives $\bar{F}_i(x_i^{(k)}) - \bar{F}_i(x_i^*) \rightarrow 0$, while the denominator converges to $c > 0$. Hence $G_i^* = 0$, and condition (3.16) forces

$$p_I = 0 \quad \text{for all } I \in M_i.$$

This constraint is necessary: if $s_i = \sum_{I \in M_i} p_I > 0$, then the axis survival function $h_i(x_i) = \bar{F}_i(x_i) - s_i G_i(\bar{F}_i(x_i))$ satisfies $\lim_{x \nearrow x_i^*} h_i(x) = h_i(x_i^*) - s_i c < h_i(x_i^*)$, so the left-limit of h_i at x_i^* is strictly less than its value, contradicting the required non-increasing property. Consequently, wherever $\bar{F}_i$ is continuous, $G_i \circ \bar{F}_i$ must be continuous as well.

- (3) If $G_i \circ \bar{F}_i$ is constant on some sub-interval while $\bar{F}_i$ is strictly decreasing there, the denominator in (3.17) vanishes for the corresponding pairs (x_i, y_i) . These pairs are excluded from the infimum by the constraint $G_i(\bar{F}_i(x_i)) > G_i(\bar{F}_i(y_i))$, so no difficulty arises.
- (4) The CDF of a G-JE random vector X_N takes the same form as the canonical expression (3.10), with each copula argument $\bar{F}_i(x_i)/\bar{F}_i(0)$ replaced by $G_i(\bar{F}_i(x_i))$.
- (5) Remark 3.4-(1) carries over to G-JE without modification. Remark 3.4-(2) extends with the additional requirement $G_i^* = \bar{F}_i(0) = 1$ for all $i \in I$, which forces those components to reduce to the canonical case. Remark 3.4-(4) carries over without modification.

Analogously to Remark 3.4-(3), the i -th axis carries zero mass if and only if equality holds in (3.16): $\sum_{I \in M_i} p_I = G_i^*$. Substituting into (3.15) gives $G_i(\bar{F}_i(x_i)) = \bar{F}_i(x_i)/G_i^*$ for all $x_i \geq 0$; evaluating at $x_i = 0$ and using $G_i(\bar{F}_i(0)) = 1$ yields $G_i^* = \bar{F}_i(0)$ and

$G_i(u) = u/\bar{F}_i(0)$, reducing component i to the canonical case. Consequently, for $n = 3$, requiring zero mass on all three axes forces $G_i^* = \bar{F}_i(0)$ for every $i \in N$, and the system $\sum_{I \in M_i} p_I = \bar{F}_i(0)$ reduces to its canonical counterpart with unique solution

$$p_I = \frac{\sum_{i \in I} \bar{F}_i(0) - \bar{F}_j(0)}{2}, \quad j \in N \setminus I,$$

valid when $\{\bar{F}_i(0)\}_{i \in N}$ satisfies the triangle inequality (see Remark 3.4-(3)).

The analog of Remark 3.4-(5), however, does not extend to G -JE. In the canonical setting, the tight condition $\sum_{i \in N} \bar{F}_i(0) = n - 1$ implies $(n - 1)a_0 + (n - 2) \sum_{i \in N} a_i \leq 0$, forcing all axis and origin masses to zero. For G -JE, the existence condition (3.18) becomes tight when $\sum_{i \in N} \bar{F}_i(0) - 1$ equals the right-hand side of (3.18), a quantity governed by $\{G_i^*\}$ rather than $\{\bar{F}_i(0)\}$. Since each axis carries a minimum mass of $\bar{F}_i(0) - s_i \geq \bar{F}_i(0) - G_i^* \geq 0$ regardless of the choice of $\{p_I\}$, a tight G -JE condition does not in general force the axis masses to vanish. For instance, with $n = 3$, $\bar{F}_i(0) = \frac{1}{2}$ and $G_i^* = \frac{1}{3}$ for all $i \in N$, condition (3.18) holds with equality, yet each axis carries a minimum mass of $\bar{F}_i(0) - G_i^* = \frac{1}{6} > 0$.

Analogously to Proposition 3.5, the admissible coefficients $\{p_I\}_{I \in M}$ for a trivariate G -JE random vector admit a one-parameter linear interpolation. The role of $\bar{F}_i(0)$ in the upper bounds U_I is now played by G_i^* , while the lower bounds L_I are adjusted to reflect constraint (3.16) in place of (3.4).

Proposition 3.10. *Let $n = 3$ and suppose that condition (3.18) holds. For each $I \in M$ with complementary index $j \in N \setminus I$, define*

$$p_I(\lambda) = \lambda \min_{i \in I} G_i^* + (1 - \lambda) \max \left\{ \sum_{k \in N} \bar{F}_k(0) - G_j^* - 1, 0 \right\}, \quad \lambda \in [0, 1]. \quad (3.19)$$

Then there exists $\lambda^ \in [0, 1]$ such that $\{p_I(\lambda^*)\}_{I \in M}$ satisfies both (3.16) and (3.6).*

Proof. Assume (3.18) holds and define, for each $I \in M$ with $j \in N \setminus I$,

$$U_I = \min_{i \in I} G_i^*, \quad L_I = \max \left\{ \sum_{k \in N} \bar{F}_k(0) - G_j^* - 1, 0 \right\}.$$

Here U_I is the maximum value of p_I consistent with (3.16) when all other face masses vanish, and L_I is the lower bound on p_I derived from the combined constraints (3.16) and (3.6): for $n = 3$ and $I = \{i, k\}$ with $j \in N \setminus I$, constraint (3.16) for index j gives $p_{\{i,j\}} + p_{\{k,j\}} \leq G_j^*$; substituting into (3.6) yields $p_I \geq \sum_{l \in N} \bar{F}_l(0) - G_j^* - 1$, hence L_I as stated.

The inequality $U_I \geq L_I$ holds by condition (3.18): for any $j \in N \setminus I$,

$$\min_{i \in I} G_i^* + G_j^* \geq \sum_{l \in N} G_l^* - \max_{l \in N} G_l^* \geq \sum_{l \in N} \bar{F}_l(0) - 1,$$

where the first inequality follows because $\max_l G_l^* \geq G_l^*$ for every l , and the second from (3.18); rearranging gives $U_I \geq \sum_{l \in N} \bar{F}_l(0) - G_j^* - 1$.

Substituting $p_I(\lambda) = \lambda U_I + (1 - \lambda) L_I$ into (3.16) and (3.6) and solving for λ yields the requirement

$$\ell \leq \lambda \leq u_i \quad \text{for all } i \in N, \quad (3.20)$$

where

$$\ell = \max \left\{ \frac{\sum_{k \in N} \bar{F}_k(0) - 1 - \sum_{I \in M} L_I}{\sum_{I \in M} (U_I - L_I)}, 0 \right\}, \quad u_i = \frac{G_i^* - \sum_{I \in M_i} L_I}{\sum_{I \in M_i} (U_I - L_I)}.$$

We show that $0 \leq \ell \leq u_i \leq 1$ for every $i \in N$.

Non-negativity of u_i . For fixed $i \in N$, the numerator $G_i^* - \sum_{I \in M_i} L_I$ takes one of the following values according to how many terms in $\sum_{I \in M_i} L_I$ are strictly positive: G_i^* (both zero); $G_i^* + G_j^* - (\sum_k \bar{F}_k(0) - 1)$ for $j \in N \setminus \{i\}$ (exactly one positive); or $\sum_k G_k^* - 2(\sum_k \bar{F}_k(0) - 1)$ (both positive). All three are non-negative by condition (3.18): the first trivially; the second since $\sum_k \bar{F}_k(0) - 1 \leq \sum_k G_k^* - \max_k G_k^* \leq G_i^* + G_j^*$ for any $j \neq i$; the third since $\sum_k \bar{F}_k(0) - 1 \leq$

$\frac{1}{2} \sum_k G_k^*$ (first term of (3.18) with $n = 3$). Hence $u_i \geq 0$.

Upper bound $u_i \leq 1$. The bound $u_i \leq 1$ is equivalent to $G_i^* \leq \sum_{I \in M_i} U_I = \min(G_i^*, G_j^*) + \min(G_i^*, G_k^*)$ for $\{j, k\} = N \setminus \{i\}$. This sum falls below G_i^* only if $G_i^* > G_j^* + G_k^*$, a strict dominance condition that can hold for at most one index i simultaneously. Hence $\min_{i \in N} u_i \leq 1$.

Non-emptiness: $\ell \leq \min_{i \in N} u_i$. Since $\sum_{I \in M_i} (U_I - L_I) \leq \sum_{I \in M} (U_I - L_I)$ and the numerator of u_i is non-negative, $u_i \geq (G_i^* - \sum_{I \in M_i} L_I) / \sum_{I \in M} (U_I - L_I)$. By definition, $L_{N \setminus \{i\}} = \max\{\sum_k \bar{F}_k(0) - G_i^* - 1, 0\} \geq \sum_k \bar{F}_k(0) - G_i^* - 1$, so $\sum_k \bar{F}_k(0) - G_i^* - 1 - L_{N \setminus \{i\}} \leq 0$. Writing $\sum_{I \in M} L_I = \sum_{I \in M_i} L_I + L_{N \setminus \{i\}}$ and adding the non-positive term $\sum_k \bar{F}_k(0) - G_i^* - 1 - L_{N \setminus \{i\}}$ to the numerator:

$$u_i \geq \frac{G_i^* - \sum_{I \in M_i} L_I + \sum_k \bar{F}_k(0) - G_i^* - 1 - L_{N \setminus \{i\}}}{\sum_{I \in M} (U_I - L_I)} = \frac{\sum_k \bar{F}_k(0) - 1 - \sum_{I \in M} L_I}{\sum_{I \in M} (U_I - L_I)}.$$

Since this lower bound equals the untruncated part of ℓ , we have $u_i \geq \ell$ for every $i \in N$.

Therefore λ^* exists in $[\ell, \min_{i \in N} u_i] \subseteq [0, 1]$. $\square$

In the next example, we illustrate several basic choices of the functions G_i and the rich G-JE structures they generate.

Example 3.11.

(1) Consider the linear functions

$$G_i(u) = \frac{u - a_i}{b_i - a_i}, \quad u \in [a_i, b_i],$$

extended by $G_i(u) = 0$ for $u \in [0, a_i]$ and $G_i(u) = 1$ for $u \in [b_i, 1]$. The boundary conditions $G_i(a_i) = 0$ and $G_i(b_i) = 1$ are satisfied by construction, and uniquely determine the slope $1/(b_i - a_i)$. Since G_i is linear on $[a_i, b_i]$, the ratio $(\bar{F}_i(x_i) - \bar{F}_i(y_i)) / (G_i(\bar{F}_i(x_i)) - G_i(\bar{F}_i(y_i)))$ equals $b_i - a_i$ whenever both $\bar{F}_i(x_i), \bar{F}_i(y_i) \in [a_i, b_i]$, and is larger otherwise.

Hence $G_i^* = b_i - a_i$, provided $\bar{F}_i$ attains at least one value in (a_i, b_i) . Setting $a_i = 0$ and $b_i = \bar{F}_i(0)$ gives $G_i(u) = u/\bar{F}_i(0)$ and $G_i^* = \bar{F}_i(0)$, recovering the canonical JE.

Now take $a_i > 0$ and $b_i = \bar{F}_i(0)$, and assume each $\bar{F}_i$ attains at least one value in $[a_i, \bar{F}_i(0))$. Then $G_i^* = \bar{F}_i(0) - a_i$, and condition (3.18) becomes

$$\sum_{i \in N} \bar{F}_i(0) - 1 \leq \min \left\{ \frac{n-2}{n-1} \sum_{i \in N} (\bar{F}_i(0) - a_i), \sum_{i \in N} (\bar{F}_i(0) - a_i) - \max_{i \in N} (\bar{F}_i(0) - a_i) \right\}.$$

Writing $\bar{F}_i(0) - a_i = G_i^*$ and expanding each case:

Case 1 (symmetric regime, first term binding): the condition reduces to

$$\sum_{i \in N} \bar{F}_i(0) + (n-2) \sum_{i \in N} a_i \leq n-1.$$

Compared with (3.1), the extra term $(n-2) \sum_{i \in N} a_i$ reflects the additional restriction imposed by truncating the face distributions at a_i from below.

Case 2 (asymmetric regime, second term binding): the condition becomes

$$\sum_{i \in N} a_i + \max_{i \in N} (\bar{F}_i(0) - a_i) \leq 1,$$

a constraint that primarily controls the balance between the lower bounds $\{a_i\}$ and the largest effective face mass.

Setting $a_i = 1 - q_i$ for parameters $q_i \in [F_i(0), 1]$ (so that $G_i^* = q_i - F_i(0) \geq 0$) and using $\bar{F}_i(0) = 1 - F_i(0)$, these two conditions may equivalently be written as

$$\begin{aligned} \sum_{i \in N} (1 - q_i) + \frac{\sum_{i \in N} (q_i - F_i(0))}{n-1} &\leq 1, \\ \sum_{i \in N} (1 - q_i) + \max_{i \in N} (q_i - F_i(0)) &\leq 1. \end{aligned}$$

The second inequality is precisely the necessary and sufficient condition for the existence of the Tail-ME random vector with probability vector $(q_1, \dots, q_n)$ studied in [Cheung et al. \(2017\)](#). The first inequality additionally ensures that the G-JE variant concentrates its face mass on the lower portion of the box $B = \{x_N \in [0, \infty)^n : 0 \leq x_i \leq F_i^{-1}(q_i)\}$, rather than over the full n -cube B .

As a concrete illustration, modify Example 3.6 by setting $a_i = 1/8$ (and retaining $b_i = \bar{F}_i(0) = 1/2$) for all $i \in N$. Then $G_i^* = 3/8$, and one verifies that (3.18) holds. The transformed argument is

$$G_i(\bar{F}_i(x_i)) = \frac{(1 - x_i)/2 - 1/8}{3/8} = (1 - \frac{4}{3}x_i)_+, \quad x_i \in [0, 1],$$

where $(t)_+ = \max\{t, 0\}$ denotes the positive part. Applying Proposition 3.10 with $U_I = 3/8$ and $L_I = 1/8$ for every $I \in M$ gives

$$p_I(\lambda) = \frac{3}{8}\lambda + \frac{1}{8}(1 - \lambda) = \frac{\lambda}{4} + \frac{1}{8}, \quad \lambda \in [\frac{1}{6}, \frac{1}{4}].$$

Retaining the copulas C_{12} , C_{13} , C_{23} from Example 3.6 and substituting the transformed arguments, the CDF of X_N is

$$F_N(x_N) = \begin{cases} 1 - \frac{1}{2} \sum_{i \in N} (1 - x_i) \\ \quad + \left(\frac{\lambda}{4} + \frac{1}{8}\right) \left[(1 - \frac{4}{3}x_1)_+ (1 - \frac{4}{3}x_2)_+ \right. \\ \quad \quad \left. + \min \left\{ (1 - \frac{4}{3}x_1)_+, (1 - \frac{4}{3}x_3)_+ \right\} \right. \\ \quad \quad \left. + \max \left\{ (1 - \frac{4}{3}x_2)_+ + (1 - \frac{4}{3}x_3)_+ - 1, 0 \right\} \right], & x_N \in [0, 1]^3, \\ 0, & \text{otherwise.} \end{cases}$$

The effective face support is confined to $[0, 3/4)$ in each active coordinate, strictly smaller than the $[0, 1]$ range of the canonical JE; the support of X_N is depicted in Figure 3.

Support of G-JE X_N

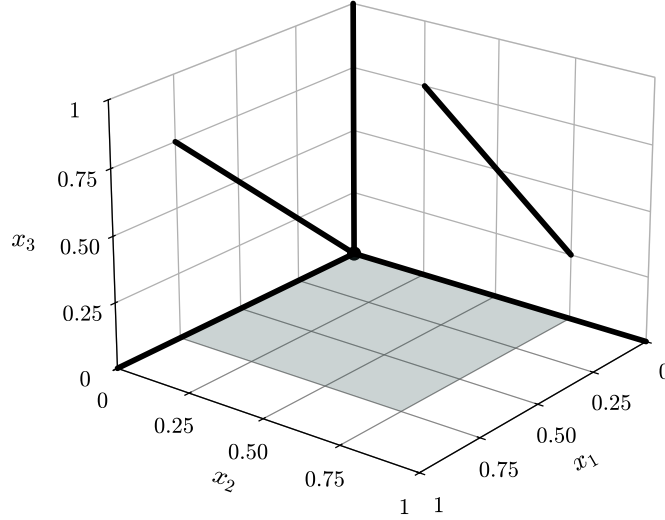


Figure 3: Support of the G-JE random vector X_N with linear truncation functions G_i at level $a_i = 1/8$, restricting the effective face support to $[0, 3/4)$ in each active coordinate.

(2) Consider the power functions

$$G_i(u) = \left(\frac{u - a_i}{b_i - a_i} \right)^\gamma, \quad u \in [a_i, b_i],$$

with exponent $\gamma > 0$, extended by $G_i(u) = 0$ for $u \in [0, a_i]$ and $G_i(u) = 1$ for $u \in [b_i, 1]$.

This recovers the linear case at $\gamma = 1$. For general γ , the ratio in (3.17) equals $(b_i - a_i)^\gamma (u - v) / (u^\gamma - v^\gamma)$ (with $u = s + a_i$, $v = t + a_i$, $s > t \geq 0$). By the mean value theorem, $(u^\gamma - v^\gamma) / (u - v) = \gamma \xi^{\gamma-1}$ for some $\xi \in (v, u)$, so the ratio equals $(b_i - a_i)^\gamma / (\gamma \xi^{\gamma-1})$.

Case $\gamma \in (0, 1)$: since $\gamma - 1 < 0$, the ratio vanishes as $\xi \rightarrow 0^+$, which occurs when $v \rightarrow a_i^+$. Hence $G_i^* = 0$ whenever $\bar{F}_i$ attains values in $(a_i, a_i + \varepsilon)$ for some $\varepsilon > 0$. Condition (3.18) then reduces to $\sum_{i \in N} \bar{F}_i(0) \leq 1$, the ME condition (1.2): the strong concavity of G_i near

a_i forces all face probabilities to vanish, recovering ME as the only feasible structure.

Case $\gamma > 1$: the ratio is minimised as $\xi \rightarrow b_i$ (since $\gamma - 1 > 0$ and $\xi^{\gamma-1}$ is increasing), which occurs as $v \rightarrow u^-$ with $u \rightarrow b_i$. Setting $a_i = 0$ and $b_i = \bar{F}_i(0)$ and assuming $\bar{F}_i$ attains values in $(b_i - \varepsilon, b_i)$ for some $\varepsilon > 0$, one obtains $G_i^* = \bar{F}_i(0)/\gamma$. Condition (3.18) then becomes

$$\sum_{i \in N} \bar{F}_i(0) \leq \min \left\{ \frac{\gamma(n-1)}{\gamma(n-1) - (n-2)}, \frac{\gamma - \max_{i \in N} \bar{F}_i(0)}{\gamma - 1} \right\}.$$

Both terms are decreasing in γ from the canonical JE bound of $n-1$ (at $\gamma = 1^+$) toward 1 as $\gamma \rightarrow \infty$, confirming that larger exponents impose progressively stronger constraints relative to the canonical case.

Another extension to JE comes as an m -exclusivity hierarchy in which JE corresponds to the least restrictive level. Strengthening the exclusion condition of JE to smaller sub-groups of components yields a nested family of conditions interpolating between JE and ME.

Definition 3.12. Let $2 \leq m \leq n$. A non-negative random vector X_N is said to be m -exclusive if

$$\bar{F}_I(0_I) = 0, \text{ or equivalently, } \prod_{i \in I} X_i = 0 \text{ almost surely, for all } I \subseteq N \text{ with } |I| = m,$$

that is, no m components are simultaneously strictly positive almost surely. The special cases $m = n$ and $m = 2$ recover JE and ME, respectively.

Since $\mathbb{P}(X_i > 0, i \in J) \leq \mathbb{P}(X_i > 0, i \in I)$ for any $I \subseteq J$, m -exclusivity immediately implies d -exclusivity for every $d \geq m$: any set of $d > m$ components contains a sub-group of size m with zero joint positivity probability, hence itself has zero joint positivity probability. Consequently,

the m -exclusivity conditions form a decreasing chain:

$$\text{ME} = 2\text{-exclusive} \Rightarrow 3\text{-exclusive} \Rightarrow \cdots \Rightarrow n\text{-exclusive} = \text{JE}.$$

Theorem 3.13. *Let $F_1, \dots, F_n$ be marginal distributions on $[0, \infty)$ and let $2 \leq m \leq n$. An m -exclusive random vector X_N with these marginals exists if and only if*

$$\sum_{i \in N} \bar{F}_i(0) \leq m - 1. \quad (3.21)$$

Proof. Necessity. If X_N is m -exclusive, then no m components are simultaneously positive, so $\sum_{i \in N} \mathbf{1}_{\{X_i > 0\}} \leq m - 1$ almost surely. Taking expectations gives $\sum_{i \in N} \bar{F}_i(0) \leq m - 1$.

Sufficiency. Assume (3.21) holds. The proof proceeds exactly as that of Theorem 3.1, with the face collection M replaced by $M^{(m)} = \{I \subset N : 2 \leq |I| \leq m - 1\}$. Restricting mass to faces of size at most $m - 1$ ensures that no m components are ever simultaneously positive, thereby enforcing m -exclusivity by construction.

The feasibility argument (Steps 1–4 of the proof of Theorem 3.1) applies to the linear program with face collection $M^{(m)}$. The dual constraint $\sum_{i \in I} r_i \geq |I| - 1$ for $I \in M^{(m)}$ is satisfied at the symmetric vertex $r^* = ((m-2)/(m-1), \dots, (m-2)/(m-1))$, since $|I|(m-2)/(m-1) \geq |I| - 1$ holds for all $|I| \leq m - 1$, and at each permutation of $(1, \dots, 1, 0)$. By strong duality, the maximum of $\sum_{I \in M^{(m)}} (|I| - 1) p_I$ equals

$$\min \left\{ \frac{m-2}{m-1} \sum_{i \in N} \bar{F}_i(0), \sum_{i \in N} \bar{F}_i(0) - \max_{i \in N} \bar{F}_i(0) \right\}.$$

In both cases, this minimum is at least $\sum_{i \in N} \bar{F}_i(0) - 1$ under condition (3.21): the second term yields the trivial bound $\max_{i \in N} \bar{F}_i(0) \leq 1$, while the first requires $\sum_{i \in N} \bar{F}_i(0) \leq m - 1$. $\square$

Corollary 3.14. *Let $F_1, \dots, F_n$ be marginal distributions on $[0, \infty)$, let $2 \leq m \leq n$, and let*

$G_1, \dots, G_n$ be functions as in Definition 3.7. A G m -exclusive random vector X_N with marginals $F_1, \dots, F_n$ exists if and only if

$$\sum_{i \in N} \bar{F}_i(0) - 1 \leq \min \left\{ \frac{m-2}{m-1} \sum_{i \in N} G_i^*, \sum_{i \in N} G_i^* - \max_{i \in N} G_i^* \right\}. \quad (3.22)$$

Proof. The proof follows the argument of Theorem 3.8, applied to the restricted face collection $M^{(m)} = \{I \subset N : 2 \leq |I| \leq m-1\}$ in place of M , with G_i^* in place of $\bar{F}_i(0)$ in the linear program, and with m taking the role of n throughout the dual analysis. $\square$

Remark 3.15. *The m -exclusivity hierarchy admits a transparent interpretation within the canonical JE framework.*

Connection to face probabilities. A canonical JE random vector X_N is m -exclusive if and only if $p_I = 0$ for every $I \in M$ with $|I| \geq m$. For the forward direction: if X_N is m -exclusive and $J \in M$ with $|J| \geq m$, choose any $I \subseteq J$ with $|I| = m$; then

$$p_J \leq \sum_{\substack{K \in M \\ K \supseteq I}} p_K = \mathbb{P}(X_i > 0, i \in I) = 0.$$

Conversely, if $p_J = 0$ for all $|J| \geq m$, then for any $I \subseteq N$ with $|I| = m$:

$$\mathbb{P}(X_i > 0, i \in I) = \sum_{\substack{K \in M \\ K \supseteq I, |K| \geq m}} p_K = 0,$$

confirming m -exclusivity. As m decreases from n to 2, faces of successively larger dimension are suppressed: at $m = n$ the full canonical JE support is recovered; at $m = 2$ all faces are zero ($p_I = 0$ for all $I \in M$), consistent with Remark 3.4-(1), and mass concentrates entirely on axes and origin, recovering ME.

Parallel structural properties. The results of Remark 3.4 extend to m -exclusive vectors with M replaced by $M^{(m)}$ and (3.1) replaced by (3.21). Items (1)–(4) carry over without modification.

For item (5), suppose $\sum_{i \in N} \bar{F}_i(0) = m - 1$. Summing (3.3) over all $i \in N$ at $x_i = 0$ with $M^{(m)}$ in place of M , and using $|I| \leq m - 1$ for every $I \in M^{(m)}$, the same argument as Remark 3.4-(5) yields

$$(m - 1) c_0 + (m - 2) \sum_{i \in N} c_i \leq 0,$$

where $c_0 = \mathbb{P}(X_i = 0, i \in N)$ and $c_i = \mathbb{P}(X_i > 0; X_j = 0, j \neq i)$. For $m \geq 3$, both coefficients $m - 1 \geq 2$ and $m - 2 \geq 1$ are strictly positive, forcing $c_0 = 0$ and $c_i = 0$ for every $i \in N$; mass concentrates entirely on $M^{(m)}$ with $\sum_{I \in M^{(m)}} p_I = 1$. For $m = 2$ (ME), the coefficient of $\sum_i c_i$ vanishes and only $c_0 = 0$ is implied, with axis masses remaining unrestricted — consistent with the ME structure. The G m -exclusive case follows identically via Corollary 3.14, with $\bar{F}_i(0)$ replaced by G_i^* throughout.

Parametric families. Proposition 3.5 provides a one-parameter family for the trivariate JE ($n = m = 3$). For $n = 3$ the hierarchy collapses to just ME ($m = 2$) and JE ($m = 3$), with no non-trivial intermediate level. Non-trivial intermediate levels first arise at $n = 4$ with $m = 3$: here $|M^{(3)}| = 6$ unknowns are subject to 4 constraints from (3.4) and 1 from (3.6), so the feasible polytope generically has dimension greater than one and the one-parameter representation of Proposition 3.5 does not extend. The LP feasibility argument of Theorem 3.13 nonetheless guarantees existence of at least one admissible solution under (3.21), with full copula freedom within each admissible face.

A natural further refinement allows specific faces of size $m - 1$ to be deactivated while retaining others, yielding a finer hierarchy governed by both the global condition (3.21) and local sub-vector conditions.

4 Further Discussion

The preceding sections developed a comprehensive theory of JE and its extensions. We characterised the admissible marginals, derived an explicit canonical construction, introduced the G-JE generalisation via marginal distortion functions, and established the m -exclusivity hierarchy interpolating between ME and JE. We now address four complementary aspects that complete the picture.

4.1 Threshold and reflected extensions

The primary variant studied throughout is *JE from below*, defined for non-negative X_N by $\prod_{i \in N} X_i = 0$ almost surely. A symmetric notion applies to non-positive random vectors: Y_N is *JE from above* if $\mathbb{P}(Y_i < 0, i \in N) = 0$, i.e. no n components are simultaneously strictly negative. Setting $X_i = -Y_i \geq 0$, Y_N is JE from above if and only if X_N is JE from below, and condition (3.1) becomes

$$\sum_{i \in N} \mathbb{P}(Y_i < 0) \leq n - 1.$$

All results of Sections 2–3 transfer immediately by this reflection. The same argument applies at every level of the m -exclusivity hierarchy: a non-positive vector Y_N is *m-exclusive from above* if $\mathbb{P}(Y_i < 0, i \in I) = 0$ for all $I \subseteq N$ with $|I| = m$, which via $X_i = -Y_i$ is equivalent to X_N being *m-exclusive from below* with existence condition $\sum_{i \in N} \mathbb{P}(Y_i < 0) \leq m - 1$.

These cases are unified by a threshold formulation. For thresholds $l_N = (l_i, i \in N) \in \mathbb{R}^n$, a random vector Z_N is *m-exclusive above level l_N* if

$$\mathbb{P}(Z_i > l_i, i \in I) = 0 \quad \text{for all } I \subseteq N \text{ with } |I| = m.$$

The excess vector $W_i = (Z_i - l_i)_+ \geq 0$ is then *m-exclusive from below* with marginal tail probability $\bar{F}_{W_i}(0) = \mathbb{P}(Z_i > l_i)$, and the existence condition for Z_N with prescribed marginals

$\{F_{Z_i}\}$ is

$$\sum_{i \in N} \mathbb{P}(Z_i > l_i) \leq m - 1. \quad (4.1)$$

All canonical and G-constructions of Section 3 apply directly to W_N and transfer to Z_N via the shift $Z_i = W_i + l_i$. The case $m = n$ with $l_N = 0_N$ and $Z_N \geq 0$ recovers JE from below; the reflected variant (JE from above on $Y_N \leq 0$) is handled via $Z_i = -Y_i$ with $l_N = 0_N$, as described above, rather than directly through the threshold formulation. Setting $m = 2$ and $l_i = F_{Z_i}^{-1}(q_i)$ for parameters $q_i \in (0, 1)$, condition (4.1) reduces to $\sum_{i \in N} (1 - q_i) \leq 1$, the existence condition for ME on the excesses $W_i = (Z_i - l_i)_+$. The corresponding G-variant (Corollary 3.14) reduces precisely to the existence condition for the Tail-ME random vector with probability vector $(q_1, \dots, q_n)$ studied in Cheung et al. (2017), as established in Example 3.11-(1).

4.2 Aggregate risk and stop-loss implications

A central motivation for extremal dependence modelling in insurance mathematics is the derivation of distributional bounds on aggregate losses. For a random vector X_N with fixed non-negative marginals $F_1, \dots, F_n$, the aggregate $S = \sum_{i \in N} X_i$ drives excess-of-loss premiums and solvency capital requirements. Its variance decomposes as

$$\text{Var}(S) = \sum_{i \in N} \text{Var}(X_i) + 2 \sum_{\substack{i, j \in N \\ i < j}} \text{Cov}(X_i, X_j),$$

where $\text{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j]$. Since the marginals are held fixed, $\mathbb{E}[X_i]$ and $\text{Var}(X_i)$ are invariant across dependence structures, and the variance of S is determined entirely by the pairwise covariances. Under ME, $X_i X_j = 0$ almost surely for every $i \neq j$, so $\mathbb{E}_{ME}[X_i X_j] = 0$ and

$$\text{Cov}_{ME}(X_i, X_j) = -\mathbb{E}[X_i] \mathbb{E}[X_j] \leq 0,$$

giving

$$\text{Var}(S_{ME}) = \sum_{i \in N} \text{Var}(X_i) - 2 \sum_{\substack{i, j \in N \\ i < j}} \mathbb{E}[X_i] \mathbb{E}[X_j] \leq \sum_{i \in N} \text{Var}(X_i).$$

Under any m -exclusive structure with $m \geq 3$, the pairwise constraint is lifted: no two components are required to satisfy $X_i X_j = 0$ almost surely, and since $X_i, X_j \geq 0$ one has $\mathbb{E}[X_i X_j] \geq 0$ for all $i \neq j$. The difference in aggregate variances between any m -exclusive ($m \geq 3$) and ME distributions with the same marginals is therefore

$$\text{Var}(S) - \text{Var}(S_{ME}) = 2 \sum_{\substack{i, j \in N \\ i < j}} (\text{Cov}(X_i, X_j) - \text{Cov}_{ME}(X_i, X_j)) = 2 \sum_{\substack{i, j \in N \\ i < j}} \mathbb{E}[X_i X_j] \geq 0, \quad (4.2)$$

where the second equality uses $\mathbb{E}_{ME}[X_i X_j] = 0$ and the inequality follows from $X_i, X_j \geq 0$. Relaxing ME to any m -exclusive structure with $m \geq 3$ therefore weakly increases aggregate variance for fixed marginals, with equality if and only if $\mathbb{E}[X_i X_j] = 0$ for all $i \neq j$, i.e. whenever the structure effectively reduces to ME. When the ME condition (1.2) fails but the JE condition (3.1) holds, the lower bound (3.6) forces $\sum_{I \in M} (|I| - 1) p_I \geq \sum_{i \in N} \bar{F}_i(0) - 1 > 0$, so at least one face E_I with $|I| \geq 2$ carries strictly positive mass $p_I > 0$. For any $i, j \in I$, $X_i > 0$ and $X_j > 0$ simultaneously on E_I , and since $\bar{F}_i(0) \geq p_I > 0$ implies $\mathbb{E}[X_i] > 0$, we obtain

$$\mathbb{E}[X_i X_j] \geq \mathbb{E}[X_i X_j \mathbf{1}_{E_I}] = p_I \mathbb{E}[X_i X_j \mid E_I] > 0,$$

where the strict inequality holds because $X_i > 0$ and $X_j > 0$ almost surely on E_I and both conditional means are strictly positive. Hence the inequality in (4.2) is strict for any JE distribution whose marginals violate the ME condition.

The m -exclusivity level also governs the higher-order joint structure of S . Under m -exclusivity, $\prod_{i \in I} X_i = 0$ almost surely for every $I \subseteq N$ with $|I| = m$; since $X_i \geq 0$, this implies $\mathbb{E}[\prod_{i \in I} X_i^{r_i}] = 0$ for any $r_i > 0$ and all $|I| \geq m$. Consequently, m -exclusivity eliminates

all joint moments of order m and above, directly constraining the cumulants of S of corresponding order. Under 3-exclusivity, all triple cross-moments $\mathbb{E}[X_i X_j X_k]$ vanish, governing the third cumulant of S in addition to the variance; under 4-exclusivity, all quartic joint cross-moments vanish as well. This systematic suppression of higher-order cross-moments as m decreases from n to 2 provides a principled and tunable mechanism for controlling the tail behaviour of the aggregate.

For excess-of-loss reinsurance, the stop-loss premium $\Pi(d) = \mathbb{E}[(S - d)_+]$ depends on the full distribution of S rather than its moments alone. When condition (1.2) holds, Cheung and Lo (2014) establishes that ME minimises $\Pi(d)$ for every retention level $d \geq 0$ among all dependence structures with the given marginals, in the sense of the convex stochastic order $S_{ME} \leq_{cx} S$. This positions ME as the natural lower benchmark for stop-loss calculations. When the ME condition fails but the JE condition (3.1) holds, ME is unavailable as a reference structure. A natural candidate for the stop-loss minimiser within the JE class is the distribution that suppresses face masses as far as constraints (3.4) and (3.6) permit, minimising $\mathbb{E}[X_i X_j]$ for each pair and approaching ME as closely as the marginals allow. Whether this candidate achieves a convex ordering within the JE class, and whether the m -exclusivity level m induces a monotone ordering of stop-loss premiums across the hierarchy for fixed marginals, are natural and actuarially important open problems.

4.3 G-JE and distortion operators

The G-JE construction has a natural interpretation in terms of distortion operators, a concept central to actuarial risk theory (Wang, 1996). For each $i \in N$, the composition $h_i = G_i \circ \bar{F}_i$ defines a valid survival function on $[0, \infty)$: it satisfies $h_i(0) = G_i(\bar{F}_i(0)) = 1$ by the boundary condition $b_i \leq \bar{F}_i(0)$; it converges to zero as $x_i \rightarrow \infty$ since $\bar{F}_i(\infty) = 0 \leq a_i$ and $G_i(0) = 0$; and it is right-continuous and non-increasing. Moreover, since $a_i \geq 0$ implies $G_i(0) = 0$, and

since $b_i \leq \bar{F}_i(0) \leq 1$ implies $G_i(1) = 1$, the function $G_i : [0, 1] \rightarrow [0, 1]$ satisfies the boundary conditions $G_i(0) = 0$ and $G_i(1) = 1$ of a distortion function in the sense of Wang (1996), monotonically transforming the survival probabilities of X_i .

Under this identification, a G-JE random vector is a canonical JE random vector in which the standardised within-face survival function $\bar{F}_i(x_i)/\bar{F}_i(0)$ of each component is replaced by the distorted counterpart $h_i(x_i) = G_i(\bar{F}_i(x_i))$. The quantity G_i^* represents the effective tail mass available for face allocation under the distorted survival function, and the existence condition (3.18) is condition (3.1) restated in terms of $\{G_i^*\}$ rather than $\{\bar{F}_i(0)\}$. The canonical JE is recovered by setting $G_i(u) = u/\bar{F}_i(0)$ on $[0, \bar{F}_i(0)]$ and $G_i(u) = 1$ for $u > \bar{F}_i(0)$, giving $G_i^* = \bar{F}_i(0)$. Concave distortions ($\gamma \in (0, 1)$ in Example 3.11-(2)) over-weight the lower tail of the within-face distribution and yield $G_i^* = 0$, causing all face probabilities to vanish; condition (3.18) then reduces to the ME condition (1.2), and the G-JE structure degenerates to ME. Convex distortions ($\gamma > 1$) under-weight the lower tail, giving $G_i^* = \bar{F}_i(0)/\gamma < \bar{F}_i(0)$ and tightening the existence condition relative to canonical JE. This connection links the G-JE framework naturally to Wang-transform premium principles and to the broader literature on distortion risk measures, suggesting potential embedding within distortion-based capital allocation models.

4.4 Connection to joint mixability

Another prominent extension of counter-monotonicity is *joint mixability* (JM) (Wang and Wang, 2016): a random vector Y_N is JM if $\sum_{i \in N} Y_i = c$ almost surely for some constant $c \in \mathbb{R}$, taken without loss of generality to be zero. Although JE and JM may appear structurally unrelated, they share the geometric feature that both supports are $(n - 1)$ -dimensional

subsets of $\mathbb{R}^n$. Formally, define

$$W_{\text{JE}} = \left\{ x_N \in [0, \infty)^n : \min_{i \in N} x_i = 0 \right\}, \quad W_{\text{JM}} = \left\{ y_N \in \mathbb{R}^n : \sum_{i \in N} y_i = 0 \right\},$$

where W_{JE} is the piecewise-linear boundary of the non-negative orthant and W_{JM} is the zero-sum hyperplane. The centering map $\Phi : W_{\text{JE}} \rightarrow W_{\text{JM}}$, defined by

$$\Phi(x_N)_i = x_i - \bar{x}, \quad i \in N, \quad \bar{x} = \frac{1}{n} \sum_{j \in N} x_j,$$

has inverse $\Phi^{-1} : W_{\text{JM}} \rightarrow W_{\text{JE}}$ given by $\Phi^{-1}(y_N)_i = y_i - \min_{j \in N} y_j$. That Φ^{-1} maps into W_{JE} follows since all components $y_i - \min_j y_j \geq 0$ and $\min_i (y_i - \min_j y_j) = 0$. Bijectivity is confirmed by direct computation: for $x_N \in W_{\text{JE}}$ (so $\min_k x_k = 0$),

$$\Phi^{-1}(\Phi(x_N))_i = (x_i - \bar{x}) - \min_{k \in N} (x_k - \bar{x}) = x_i - \bar{x} - (\min_k x_k - \bar{x}) = x_i,$$

and for $y_N \in W_{\text{JM}}$ (so $\sum_k y_k = 0$),

$$\Phi(\Phi^{-1}(y_N))_i = (y_i - \min_j y_j) - \frac{1}{n} \sum_{k \in N} (y_k - \min_j y_j) = y_i - \frac{1}{n} \sum_k y_k = y_i.$$

Since Φ is affine and Φ^{-1} is continuous (as a composition of continuous functions, the min being continuous), Φ is a homeomorphism between W_{JE} and W_{JM} .

The homeomorphism induces a distributional correspondence via pushforward: if X_N is JE then $Y_N := \Phi(X_N) = (X_i - \bar{X}, i \in N)$ is JM, since $\sum_i Y_i = \sum_i X_i - n\bar{X} = 0$ almost surely; and if Y_N is JM then $X_N := \Phi^{-1}(Y_N) = (Y_i - \min_{j \in N} Y_j, i \in N)$ is JE, since $\min_{i \in N} X_i = 0$ almost surely. An important caveat is that Φ does not preserve marginals: the distribution of $Y_i = X_i - \bar{X}$ depends on the full joint law of X_N rather than solely on F_i , so the homeomorphism does not induce a correspondence between JE distributions with prescribed marginals $\{F_i\}$ and JM

distributions with a prescribed marginal structure. This observation highlights a key analytical advantage of JE over JM: JE existence is governed by the single tractable condition (3.1), whereas JM existence for prescribed marginals requires substantially more complex analytical conditions; see Wang and Wang (2016) for a detailed treatment.

For canonical JE random vectors, the homeomorphism carries additional algebraic structure. On face E_I , where $X_j = 0$ for $j \in N \setminus I$ and $X_i > 0$ for $i \in I$, the image under Φ satisfies

$$\Phi(x_N)_j = -\frac{1}{n} \sum_{k \in I} x_k \quad \text{for all } j \in N \setminus I,$$

so all components indexed by $N \setminus I$ are equal on $\Phi(E_I)$. This block-equality structure in the JM image of each canonical JE face facilitates explicit computation of JM distributions arising from the canonical construction and suggests that the bijection may be exploited to transfer distributional results between the two frameworks. Figure 4 illustrates W_{JE} , W_{JM} , and the action of Φ for $n = 3$.

5 Conclusions

This paper introduces joint exclusivity (JE) as a tractable framework for extremal negative dependence in arbitrary dimensions. By replacing the pairwise constraint of mutual exclusivity with the global requirement that not all n components are simultaneously positive, JE yields a strictly richer class of dependence structures while retaining a complete and explicit characterisation: a JE random vector with prescribed marginals $F_1, \dots, F_n$ exists if and only if $\sum_{i \in N} \bar{F}_i(0) \leq n - 1$, a condition whose gap from the mutual exclusivity bound $\sum_{i \in N} \bar{F}_i(0) \leq 1$ widens with dimension. The canonical construction separates the allocation of probability mass across the faces of the support—governed by a finite-dimensional linear system in the face-mass coefficients—from the specification of within-face dependence through freely chosen copulas. In

Supports of JE and JM

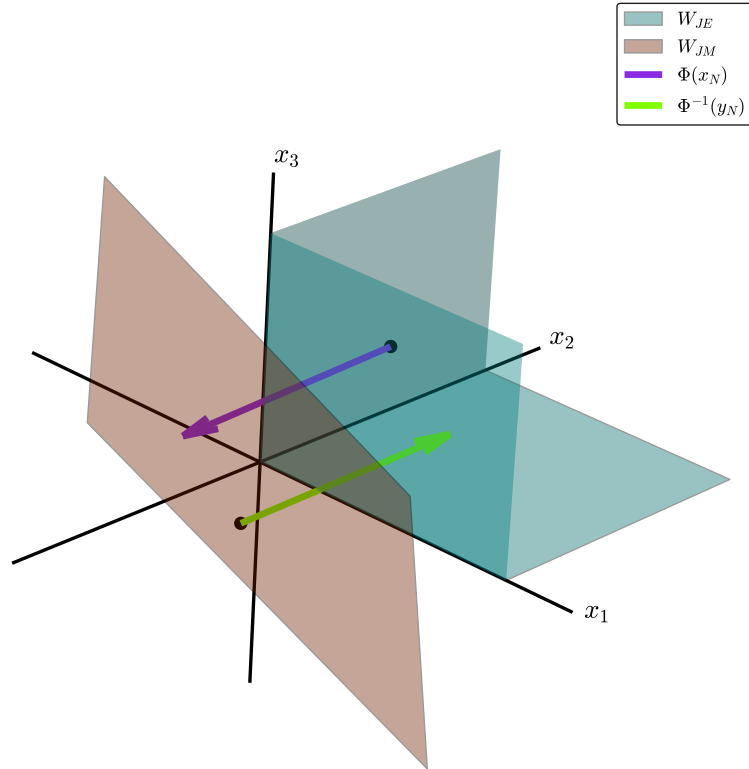


Figure 4: The support sets W_{JE} and W_{JM} for JE and JM random vectors ($n = 3$), and the action of the homeomorphism Φ .

the trivariate case this system reduces to a one-parameter family, yielding a particularly transparent and implementable model.

Two extensions enrich the framework further. The generalised JE (G-JE) replaces the standardised within-face survival functions by distorted counterparts through Wang-type distortion operators G_i , accommodating restricted or reshaped face supports and connecting JE to distortion-based premium principles. The m -exclusivity hierarchy, parametrised by $2 \leq m \leq n$, interpolates between mutual exclusivity ($m = 2$) and JE ($m = n$), each level characterised by the single inequality $\sum_{i \in N} \bar{F}_i(0) \leq m - 1$ and admitting a corresponding G-variant. Together, these extensions provide a unified and flexible toolkit for specifying negative dependence across a wide range of modelling contexts.

Several directions emerge naturally for future work. Chief among these is the stop-loss ordering of JE aggregates: while the variance of $\sum_{i \in N} X_i$ under JE exceeds that under ME for fixed marginals, a complete characterisation of excess-of-loss premiums within the JE class remains open and would be of direct relevance to reinsurance pricing. Equally important are statistical questions, including the estimation of face-mass coefficients from data and the development of goodness-of-fit procedures for JE and G-JE models. Finally, the homeomorphism between the JE and joint mixability supports suggests that distributional results in each framework may be transferred to the other, and exploiting this correspondence in the G-JE and m -exclusive settings is a promising avenue for further unifying the theory of extremal negative dependence.

References

- Cheung, K. C., Denuit, M., and Dhaene, J. (2017). Tail mutual exclusivity and tail-var lower bounds. *Scandinavian Actuarial Journal*, 2017(1):88–104.
- Cheung, K. C., Dhaene, J., Lo, A., and Tang, Q. (2014). Reducing risk by merging counter-monotonic risks. *Insurance: Mathematics and Economics*, 54:58–65.

- Cheung, K. C. and Lo, A. (2014). Characterizing mutual exclusivity as the strongest negative multivariate dependence structure. *Insurance: Mathematics and Economics*, 55:180–190.
- Deelstra, G., Dhaene, J., and Vanmaele, M. (2011). *An Overview of Comonotonicity and Its Applications in Finance and Insurance*, pages 155–179. Springer Berlin Heidelberg.
- Dhaene, J. and Denuit, M. (1999). The safest dependence structure among risks. *Insurance: Mathematics and Economics*, 25(1):11–21.
- Dhaene, J., Denuit, M., Goovaerts, M., Kaas, R., and Vyncke, D. (2002a). The concept of comonotonicity in actuarial science and finance: applications. *Insurance: Mathematics and Economics*, 31(2):133–161.
- Dhaene, J., Denuit, M., Goovaerts, M. J., Kaas, R., and Vyncke, D. (2002b). The concept of comonotonicity in actuarial science and finance: theory. *Insurance: Mathematics and Economics*, 31(1):3–33.
- Lauzier, J.-G., Lin, L., and Wang, R. (2023). Pairwise counter-monotonicity. *Insurance: Mathematics and Economics*, 111:279–287.
- Lee, W. and Ahn, J. Y. (2014). On the multidimensional extension of countermonotonicity and its applications. *Insurance: Mathematics and Economics*, 56:68–79.
- Nelsen, R. B. (2010). *An Introduction to Copulas*. Springer Publishing Company, Incorporated.
- Puccetti, G. and Wang, R. (2015). Extremal Dependence Concepts. *Statistical Science*, 30(4):485 – 517.
- Wang, B. and Wang, R. (2016). Joint mixability. *Mathematics of Operations Research*, 41(3):808–826.

Wang, S. (1996). Premium calculation by transforming the layer premium density. *ASTIN Bulletin*, 26(1):71–92.